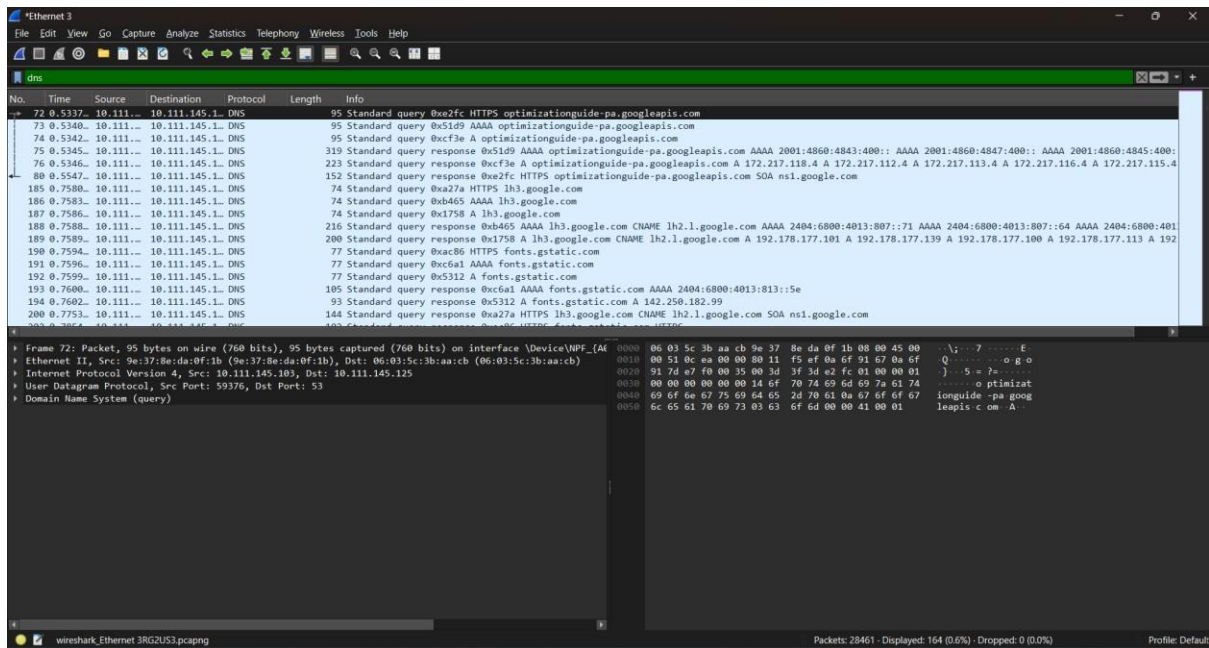


TASK -1



TCP/IP Layer	Protocol in your packet	Header fields you can mention
Application	DNS	Transaction ID, Flags, Queries
Transport	UDP	Source Port: 63514, Destination Port: 53
Internet	IPv4	Source IP: 10.111.145.103, Destination IP: 10.111.145.125
Network Access	Ethernet II	Source MAC, Destination MAC

TASK-2

OSI Model vs TCP/IP Model Comparison

OSI Layer	Model TCP/IP Model Layer	Purpose	Real-World Example
7. Application	Application	Provides network services to user applications.	Sending a WhatsApp message or browsing Google Chrome .

OSI Layer	Model TCP/IP Model Layer	Purpose	Real-World Example
6. Presentation	Application	Formats, compresses, and encrypts data.	Instagram compresses and encrypts photos before sending them.
5. Session	Application	Starts, manages, and ends communication sessions.	A WhatsApp voice call or Google Meet meeting session.
4. Transport	Transport	Ensures reliable delivery, error checking, and flow control.	YouTube video streaming uses TCP to deliver data reliably.
3. Network	Internet	Provides logical addressing (IP) and routes data between networks.	A Zomato order request is routed through the internet using IP addresses.
2. Data Link	Network Access (Link)	Transfers data between devices on the same network using MAC addresses.	Your phone connects to your Wi-Fi router using its MAC address.
1. Physical	Network Access (Link)	Transmits raw bits as electrical, optical, or wireless signals.	Data travels through Wi-Fi radio waves or an Ethernet cable .

Key Differences

OSI Model

Has **7 layers**.

Mainly a **reference model** for learning and network design.

TCP/IP Model

Has **4 layers**.

The **practical model** used on the Internet today.

OSI Model

Separates **Application, Presentation, and Session** into different layers.

Separates **Physical** and **Data Link** layers.

More detailed and easier understanding networking concepts.

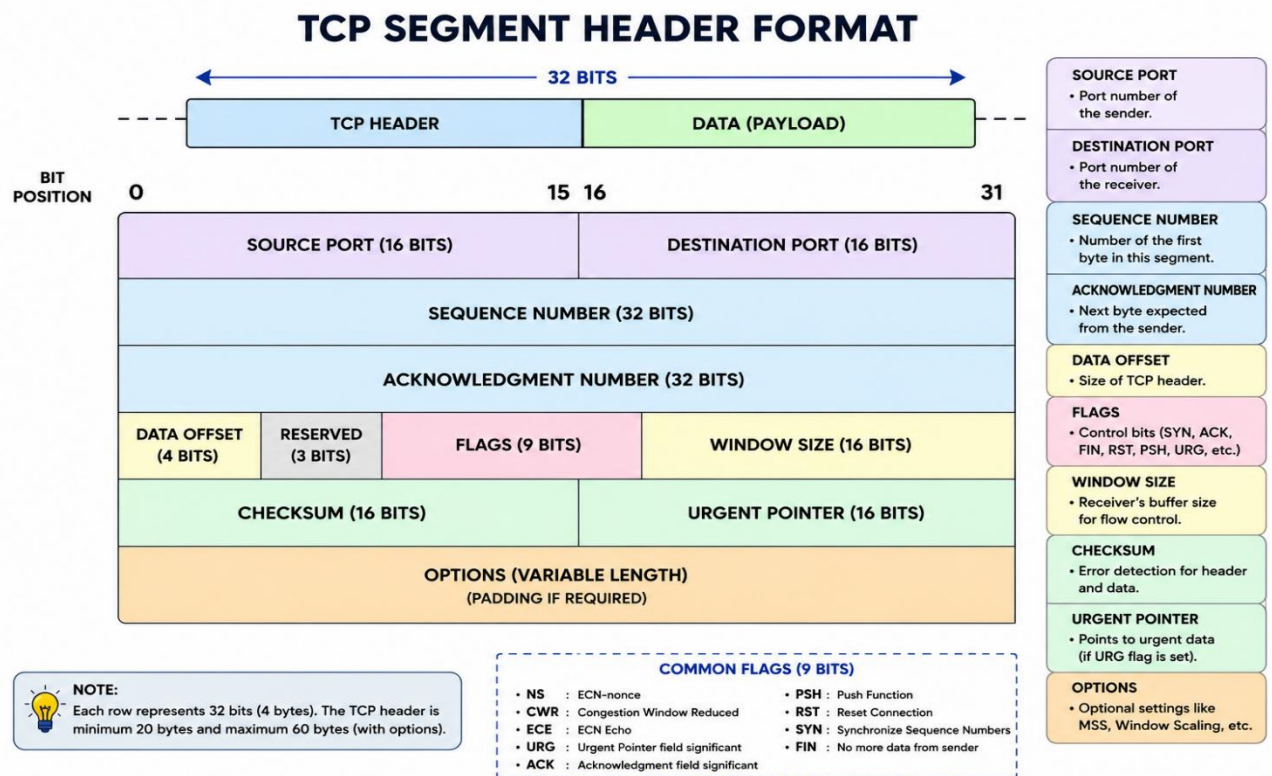
TCP/IP Model

Combines these three into a single **Application Layer**.

Combines them into the **Network Access (Link) Layer**.

Simpler and widely implemented in real-world networks.

TASK-3



TCP Header Field Function (1–2 lines)

Source Port

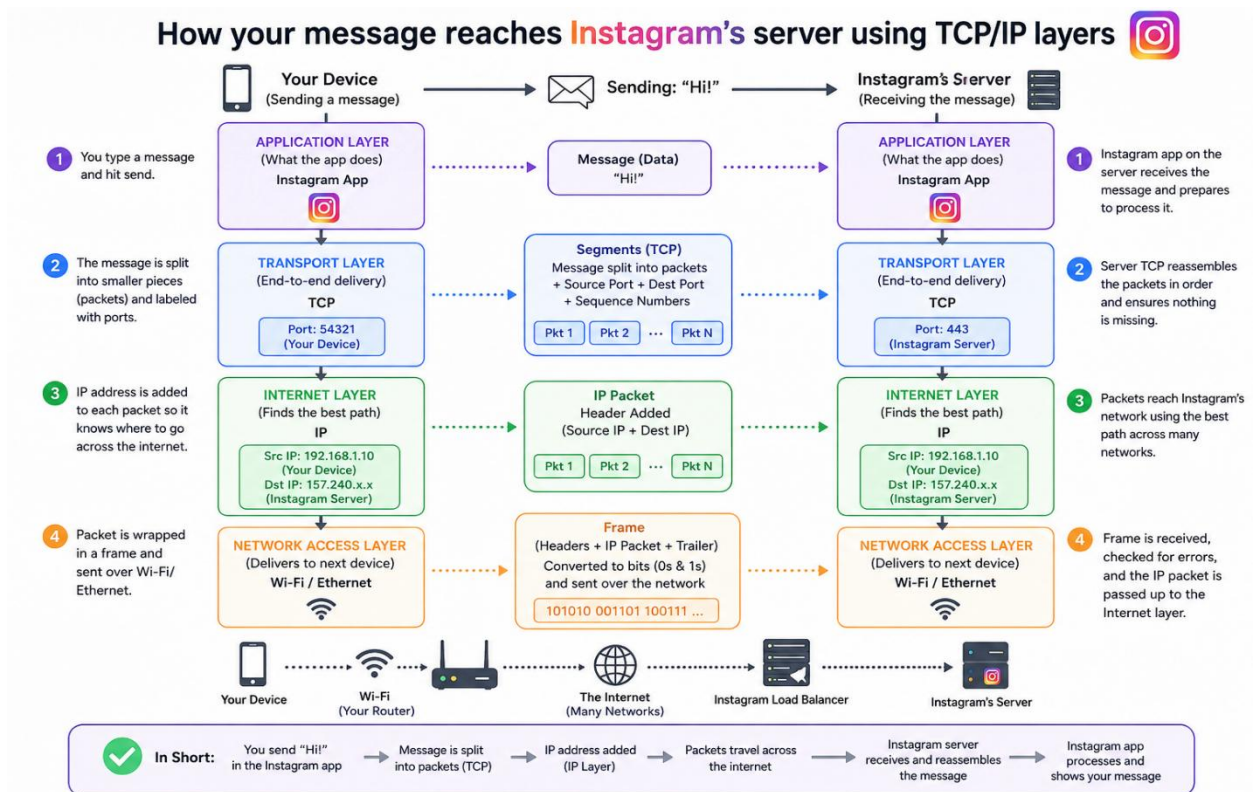
Identifies the port number of the sending application (e.g., WhatsApp, Chrome).

Destination Port

Identifies the port number of the receiving application or service (e.g., HTTP uses port 80, HTTPS uses port 443).

TCP Header Field	Function (1–2 lines)
Sequence Number	Specifies the number of the first byte in the current segment. It helps keep data in the correct order.
Acknowledgment Number	Indicates the next byte the receiver expects to receive, confirming successful receipt of previous data.
Header Length (Data Offset)	Shows the size of the TCP header so the receiver knows where the actual data begins.
Reserved	Reserved for future use and normally set to zero.
Flags (Control Bits)	Controls the connection. Examples: SYN starts a connection, ACK acknowledges data, FIN ends a connection, RST resets a connection, PSH pushes data immediately, and URG indicates urgent data.
Window Size	Indicates how much data the receiver can accept before sending another acknowledgment, helping with flow control.
Checksum	Detects errors in the TCP header and data during transmission. If errors are found, the segment is discarded.
Urgent Pointer	Points to urgent data when the URG flag is set.
Options	Provides optional features such as Maximum Segment Size (MSS), Window Scaling, or Timestamps.
Padding	Adds extra bits so the TCP header length is a multiple of 32 bits.

TASK-4



TASK-5

Here is an example analysis of a real-world network packet capture:

1. Captured Packet Overview

- **Packet Protocol:** DNS (Domain Name System Query)
- **Captured Info:** Standard query 0x1a2b A pings.bookmyshow.com

2. TCP/IP Layer Assignment

- **TCP/IP Layer:** Application Layer

3. How It Was Identified in Wireshark

- **Protocol Column:** Wireshark explicitly displays **DNS** in the *Protocol* column.
- **Packet Detail Pane:** Expanding the **Application Layer** section reveals the domain resolution request (Queries: pings.bookmyshow.com: type A, class IN).

- **Transport Protocol:** Looking one layer down in the packet details shows it uses **UDP** on **Port 53**, which is the standard transport port for standard DNS lookups.

4. Real-World App Action Represented

- **The Action:** You tap on the **BookMyShow** app on your phone or open bookmyshow.com in your web browser to check movie tickets.
- **What the Packet is Doing:** Before your phone can load the movie posters or showtimes, it needs to know the actual IP address of BookMyShow's servers. This captured DNS packet is your device asking a network domain server, *"What is the IP address for bookmyshow.com so I can connect to it?"*